

ColorWatch: Color-Tactile Interpretation to Improve Color Perception and Accessibility for People with Visual Impairments

Chung-Heon Lee

*Department of Electrical & Computer Engineering
Sungkyunkwan University
Suwon, South Korea
yi940715@g.skku.edu*

Muhammad Shahid Jabbar

*College of Information & Communication Engineering
Sungkyunkwan University
Suwon, South Korea
eeshahid@skku.edu*

Jun-Dong Cho*

*College of Information & Communication Engineering
Sungkyunkwan University
Suwon, South Korea
jdcho@skku.edu*

Abstract—This study aims toward the development of an automatic wristwatch(ColorWatch) tactile system which may assist People with Visual Impairments (PVI) to recognize object colors in their daily life. This work identified perpetual colors in daily lives and associated them with cross-modular tactile patterns. The designed system considers the fundamentals of tactile actuation, color perception, and the intuitive learnability of humans. Achromatic and monochromatic colors as well as the tone of these for each color in two levels are considered and a one-to-one cross-modular mapping with the tactile pattern has been proposed. Furthermore, the system considers artwork as well as a variety of general-purpose object color translation. Focus group experiments indicate good recognition results and usability of tactile patterns for the tactile actuation and their color associations. Based on the proposed concept and experimental evaluation, further ColorWatch prototype and user devices for translation of color into an automatic tactile system may be developed.

Index Terms—color recognition, haptic perception, intra-modular association, universal design, accessibility, people with visual impairment

I. INTRODUCTION

People with Visual Impairment (PVI) especially with born blindness have no intuitive perception of colors inherently. The sensory perception in the human brain is processed by the medial anterior temporal lobe (ATL) and hence, is responsible for the concrete concepts. While the dorsolateral ATL manages the abstract information, these abstract concepts lack associated sensory perception and are often associated with defining terms of any phenomena based on language and vocabulary [1]. Nevertheless, PVI can struct accurate and rich color concepts. Relevant researches indicate that by the help of

language and communication models, they not only can grasp the abstract concept of colors and difference between colors but also exhibit these concepts with properties of real-world objects including the relative similarity between yellow and orange rather than green, for example [1], [2].

Both psychological and aesthetic stimuli are involved regarding response to aspects of color in daily life. These aspects along-with emotional and physical sensations influence lifestyle, art, and commerce among many others. Symbolism, preferences, harmony, terminology, and other aspects of colors vary between universal, cultural, and individual conditioning. These color associations serve a significant role in culture, faith, art, commercial branding, and in everyday life. This demands to impart color information as comprehensive as possible to PVI in order for them to think, consider, make actions, and cause reactions more prudently [7].

Recognition of color based on non-visual stimuli for PVI has been an active research topic recently. Few schemes for art exploration have been proposed in [3], [4] but they either include voice or thermal interaction in addition to tactile recognition. [5], [6] have translated color using additional color symbols and stimuli changes. There exist multiple representative models for color spectrum [8] of which Munsell color system [9] has been incorporated in relevant researches [4] for its ease and correlation to human visual perception. Munsell color system arranges the color spectrum into three aspects of Hue, Chroma, and Value by mapping Hue onto the surface of the cylindrical coordinate system, whereas the radius of the cylinder represents chroma or saturation, and value is defined by relative height. This and other color models are sophisticated in terms of end-user learnability as well as automatic cross-modality association prototype. Therefore, we have proposed a simplified color-to-tactile translation model

This work was supported by the Korea Research Foundation Science and Technology Humanities and Social Convergence Research Project in 2019 (2018M3C1B6061353).

for PVI, incorporating all three aspects of hue, chroma, and value.

PVI have superior performance over tactile tasks and the studies show that the visual cortex area of the human brain is employed during spatial perception, tactile perception, and linguistics in PVI [13]. Tactile sensory and non-visual signals also contribute to the vision contextual modulation by accessing the neural circuits responsible for the said computations [10], [12]. The study [11] explains the cognitive aspects of multi-sensory activation and grating orientation evaluation by the sense of touch.

In this work, we introduce a novel tactile pattern and cross-modality association as a preliminary study for the design concept for ColorWatch for which extensive focus group experiment evaluations have been performed. The rest of the paper is as follows; the concept of design, selection of colors, tactile pattern, and color-tactile association are provided in section II. Experimental setup and conducted tests have been described in section III while section IV provide corresponding results. Section V concludes the paper and provides further insights for future study.

II. CONCEPT OF DESIGN

For the effectiveness of User Interface (UI) in the application, the colors most commonly used in daily life objects and products should be accommodated in design for blind people to convey through their ColorWatch. As a consequence, in this experiment, as introduced in [17], colors that are commonly used to stimulate people's purchasing sentiment, such as red, orange, yellow, green, blue, purple, pink, and black, are applied into ColorWatch. In the case of black, it is classified in the circle's point 'Ac', which indicates that they are achromatic colors. [18] also studied on buying trends of people according to the colors based on [17].

This intuitively explains the incorporation of the following colors to the ColorWatch. Also, an achromatic entry has been added in order to translate white, black, and gray colors. ColorWatch's concept design consists of two rotation enabled tactile arms which are represented here in this tactile interpretation study by a circle and a square. The circle represents detected colors or hue which are placed at 45-degree angle apart uniformly. The color composition and their respective angles of the circle are presented in Table I.

L, S, M, and D points on the second level of design are adjusted on the square shape corners. These points represent the saturation-lightness level for each color in our color space represented by the circle. Boundary level saturation is defined by saturated (S) and dark (D) on horizontal corners of the square, whereas, lighted (L) and muted (M) defines lightness level on vertical corners of the square for each color. A general depiction alongwith saturated blue color has been provided in Fig. 1. These are presented from the Berkeley Color Project (BCP) which defines the tones of the colors in the color space of BCP as in [16]. The saturation-lightness level on the square represents L, S, M, or D or each color on the circle except "Ac", for which it represents particular LSMD

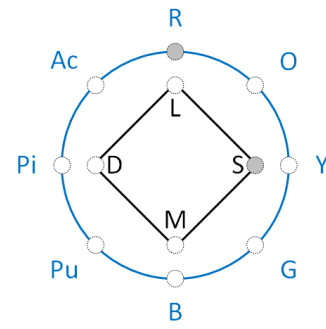


Fig. 1. ColorWatch pattern

of achromatic color in terms of white, light gray, dark gray and black. Tables II and III provide the saturation-lightness levels adopted in our study for monochromatic colors from 0-270° and achromatic (Ac) on 315° at the first layer color circle respectively. Fig. 2 provides the schematic as an example of PVI using ColorWatch during shopping.

TABLE I
COLORS ACCORDING TO THE ANGLE OF THE CIRCLE

Angle (Round)	Color
0	Red
45	Orange
90	Yellow
135	Green
180	Blue
225	Purple
270	Pink
315	Achromatic

TABLE II
SQUARE ANGLE SATURATION-LIGHTNESS LEVELS FOR MONOCHROMATIC COLORS (0-270° ON CIRCLE)

Angle (Square)	Color
0	Lighted
90	Saturated
180	Muted
270	Dark

TABLE III
SQUARE ANGLE SATURATION-LIGHTNESS LEVELS FOR ACHROMATIC COLOR (315° ON CIRCLE)

Angle (Square)	Color
0	White
90	Light Gray
180	Dark Gray
270	Black

III. USER TESTS

The experiment was conducted on a total of eight students from Sungkyunkwan University, including five male and three female students. Pre-training session on Color Watch algorithm was conducted and then 15 minutes of personal learning time was offered so that they could learn color-angle association enough individually. The pattern of the ColorWatch was printed on the TactPlus's (<https://www.tactplusprinter.com/>) swell paper with a swell printer and a total of six quizzes were given to each participant (Fig. 3). Swell printing is a technology that utilizes a special swell printer on specially-made application-specific paper by heating the paper surface which swells up to form a printed pattern that might be noticed by touching that with fingers (Fig. 4). As for the protruding part and thickness printed on the Swell paper, based on our research group interviews with a blind person, the specification for optimal recognition was set at 40mm which is the size of normal watches, and the gap was set wider and thicker than the average braille for the intuitive recognition.

In the experiment, participants were asked to wear eye masks so that they could solely rely on tactile sense for recognition, and feel the same way as blind people while solving quizzes (Fig. 5). The start and end time for each participant was measured individually by smartphone stop-watch in order to obtain time spent during recognition. The participants' answers for color recognition were noted corresponding to each tactile pattern, while blindfolded. After solving the quiz, several tests such as SUS(System Usability Scale) and NASA-TLX were conducted to determine usability, including whether it was easy to understand patterns and to learn them quickly.

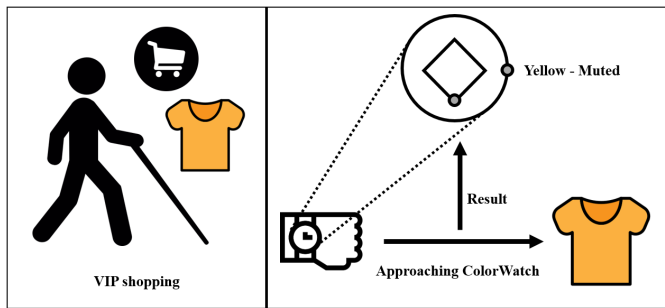


Fig. 2. PVI using ColorWatch while shopping

A. NASA-TLX Test

The NASA Task Load Index (NASA-TLX) is an evaluation developed by Human Performance Group at NASA's Ames Research Center, and it is a test that evaluates the workload of an individual while performing any task which consists of six indicators [14]. It can be used to evaluate whether a task, system, service, etc. is appropriate as per subjective mental workload assessment and has been used in various fields such as aviation, healthcare, physical education, and systems. As shown in Table IV, additional explanations are added to let experiment participants understand what exactly

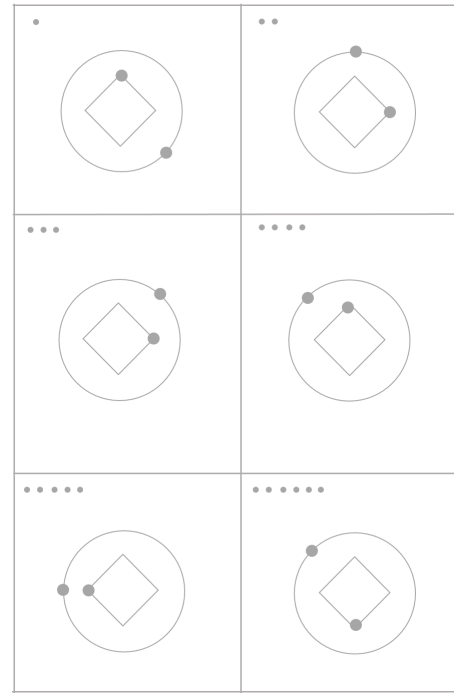


Fig. 3. Tactile quiz sample utilized for the test



Fig. 4. A printed pattern of the ColorWatch on the swell paper

they need to evaluate. Although the original test is designed for 20 points, we rated it on 10 points scale under consideration of the participants' ease familiarity with the questionnaire. Besides, the evaluation is originally conducted by additionally measuring these 6 indexes based on the received importance and correspondingly assigned weights. Contrarily, according to several studies including [19], the evaluation excluding the weighted items and the so-called Raw TLX showed better results. Under these considerations and the subjective nature of our evaluation task, we have opted Raw TLX for implementation.

TABLE IV
NASA-TLX TEST QUESTIONS

Index	Very Low << >> Very High
1. Mental Demand How much mental and perceptual activity was required? Was the task easy or demanding, simple or complex? (memory ability of pattern, color, etc)	0 1 2 3 4 5 6 7 8 9 10
2. Physical Demand How much physical activity was required? Was the task easy or demanding, slack or strenuous? (tactile cognition)	0 1 2 3 4 5 6 7 8 9 10
3. Temporal Demand How much time pressure did you feel due to the pace at which the tasks or task elements occurred? Was the pace slow or rapid?	0 1 2 3 4 5 6 7 8 9 10
4. Overall Performance How successful were you in performing the task? How satisfied were you with your performance?	0 1 2 3 4 5 6 7 8 9 10
5. Effort How hard did you have to work (mentally and physically) to accomplish your level of performance?	0 1 2 3 4 5 6 7 8 9 10
6. Frustration Level How irritated, stressed, and annoyed versus content, relaxed, and complacent did you feel during the task?	0 1 2 3 4 5 6 7 8 9 10

B. SUS (System Usability Scale)

System Usability Scale was developed by John Brooke in 1986. It is an effective way to test whether a system, service or product is useful. Without complex questions, the test can be carried out with just 10 insightful questions using a five-question Likert scale [15]. Odd-numbered questions mean a positive response while even-numbered questions mean a negative attitude. To calculate the SUS score; from odd-numbered sentences it should be subtracted 1 from the user's response, and from even-numbered sentences, it should be subtracted the corresponding value. As a result, the transformed response scale is from 0 to 4, with 4 being the most positive answer. Adding all participants' responses and multiplying the total by 2.5 will translate from 0-40 to 0-100. Below are ten corresponding questions to our experiment participants for SUS evaluations.

- 1) I think that I would like to use this service frequently.
- 2) I found the service unnecessarily complex.
- 3) I think the service was easy to use.
- 4) I think that I would need the support of a technical person to be able to use this service.
- 5) I found the various functions in this service were well integrated.
- 6) I thought there was too much inconsistency in this service.



Fig. 5. A student solving ColorWatch pattern quiz (i.e., blind-folded)

- 7) I would imagine that most people would learn to use this service very quickly.
- 8) I found the service very cumbersome to use.
- 9) I felt very confident using the service.
- 10) I needed to learn a lot of things before I could get going with this service.

IV. RESULTS

Exceeding the learnability expectations of our experimental design, participants learned the pattern faster as well as reflected a higher correct answer rate. Even though the experimental time to learn the pattern was scheduled for 15 minutes, all the participants completed the study within five minutes and averaged 95.8% correct even though it was their first time experiencing a visual blockage. In addition, it merely took slightly more than five minutes or even less than five minutes, except for one participant who had difficulty locating his fingers and figuring out the shape of the patterns. Details of each participant's time and tactile-color recognition score are explained in Table V.

A. NASA-TLX Test

The evaluations of the NASA-TLX test are provided in Table VI. It is observed that the mental demand, the physical demand, and the frustration were not much of a difficulty as the average scores of 8 participants on the tests were less than

TABLE V
ANSWER RATE AND TIME SPENT PERFORMING THE TASK

Participant	Time duration	Correct answers	Correct rate
I	10:50	6	100
J	4:05	5.5	91.67
K	5:16	6	100
L	3:50	6	100
M	3:09	5.5	91.67
N	3:50	5	83.33
O	4:14	6	100
P	3:45	6	100
		<i>Average</i>	95.83

TABLE VI
SCORE OF THE PARTICIPANTS FOR RAW NASA-TLX TEST

Index	Subjected Participant								
	I	J	K	L	M	N	O	P	Avr.
1.Mental demand	2	1	6	3	6	5	8	6	4.625
2.Physical demand	4	0	1	5	2	6	2	4	3
3.Temporal demand	3	2	3	7	2	6	8	4	4.375
4.Performance	8	5	8	10	8	6	9	9	7.875
5.Effort	5	5	3	8	6	3	5	8	5.375
6.Frustration	5	0	2	3	1	4	0	3	2.25

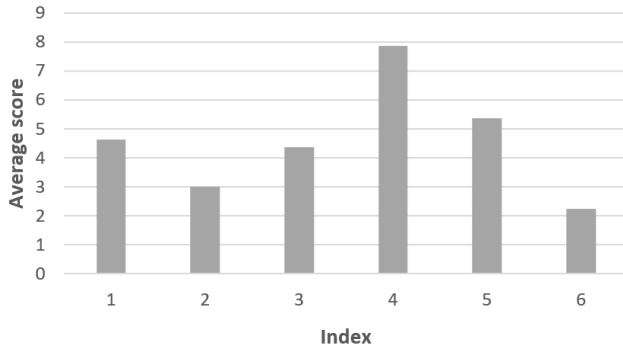


Fig. 6. Average of the NASA-TLX score of each index

five which is the median value. In the case of the performance index, it prevailed almost eight points which implies that they were generally satisfied with the performance.

As per participants feedback about how they fell regarding solving the pattern quiz, the majority of respondents said that they had difficulty identifying the protruding area because it was their first time solving those kinds of quizzes without sight. Referring to 6, 'effort (5)' and 'mental demand (1)' both still carry slightly more points on average than 'physical demand (2)', cause of that might be concluded as the requirement of more concentration on tactile perception end as their eyesight was blocked with blindfolded for the first time.

B. SUS (System Usability Scale)

In the case of this experimental test, the subjects were required to attempt the test and respond to questions without incorporating their sense of vision in order to consider PVI's restrictions. The participants had no prior experience of relying solely on tactile without a sense of vision. However, according to the SUS grading, as shown in Table VII, the average score was 70 points which exceed the threshold of the grade-B as 'Good' adjective rating (Table VIII). Especially regarding the SUS questions such as 2 ('I found the service unnecessarily complex'), 6 ('I thought there was too much inconsistency in this service') and 7 ('I would imagine that most people would learn to use this service very quickly'), the score exceeds the threshold for Grade-A 'Excellent' adjective rating (Fig. 7)

TABLE VII
SUS SCORE OF 8 PARTICIPANTS

SUS Index	Subjected Participant								
	I	J	K	L	M	N	O	P	Avr.
1	75	25	75	75	75	75	75	75	69
2	100	100	75	75	100	75	75	50	81
3	25	75	75	75	100	75	75	50	69
4	50	25	100	75	75	25	75	0	53
5	50	75	75	25	75	50	75	25	56
6	100	100	100	75	75	75	100	75	88
7	75	100	100	75	100	75	100	50	84
8	100	75	50	25	75	75	25	25	56
9	50	100	75	50	100	50	75	50	69
10	75	100	75	75	100	75	25	75	75
Avr.	70	78	80	63	88	65	70	48	70

TABLE VIII
NORMALIZED GRADES OF SUS

SUS Score	Letter grade	Adjective rating
Above 80.3	A	Excellent
Between 68 and 80.3	B	Good
68	C	Okay
Between 51 and 67	D	Poor
Below 51	F	Awful

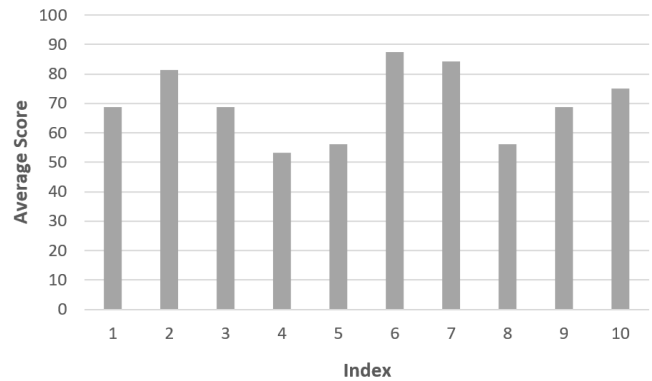


Fig. 7. Average score of participants for each SUS test index

which is a superlative score, indicating that the ColorWatch pattern is consistent, easy to understand and learn.

V. CONCLUSION

In this study, a cross-modal pattern for the color to tactile translation is introduced. For color considerations; most perpetual colors are selected for recognition along with saturation and tone interpret-ability. While the pattern is designed as per tactile actuation fundamentals for PVI sensory recognition. According to the results of the comprehensive focus group experiments, the pattern is evaluated to be consistent and easy to learn. And even though it was the first time for

the participants to solve the tactile problem, it has been demonstrated that participants took a considerably short time to learn. Therefore, it implies that the proposed ColorWatch concept with this pattern scheme is not difficult to understand for the PVI too, by the virtue of superior tactile actuation and sensory function which enables them to recognize even narrow-spaced braille better than ordinary people. These color-to-tactile association, interpretability, and learnability inspire for further development of automatic color sensory and tactile system that might aid people with visual impairments in decision making and appreciation of colors in their daily lives.

REFERENCES

- [1] E. Striem-Amit, X. Wang, Y. Bi, and A. Caramazza, "Neural representation of visual concepts in people born blind," *Nature Communications*, vol. 9, no. 1, 2018.
- [2] M. Bedny and A. Caramazza, "Perception, action, and word meanings in the human brain: the case from action verbs," *Annals of the New York Academy of Sciences*, vol. 1224, no. 1, pp. 81–95, 2011.
- [3] I. Emsley, T. Graven, N. Bird, S. Griffiths, J. Suess, and L. Shaw, "Please Touch the Art: Experiences in Developing for the Visually Impaired," *Journal of Open Research Software*, vol. 7, 2019.
- [4] J. D. I. Bartolome, L. C. Quero, J. Cho, and S. Jo, "Exploring Thermal Interaction for Visual Art Color Appreciation for the visually impaired people," *2020 International Conference on Electronics, Information, and Communication (ICEIC)*, 2020.
- [5] S. Ramsamy-Iranah, M. Maguire, J. Gardner, S. Rosunee, and N. Kistamah, "A comparison of three materials used for tactile symbols to communicate colour to children and young people with visual impairments," *British Journal of Visual Impairment*, vol. 34, no. 1, pp. 54–71, 2015.
- [6] D. Goldreich and I. M. Kanics, "Tactile Acuity is Enhanced in Blindness," *The Journal of Neuroscience*, vol. 23, no. 8, pp. 3439–3445, 2003.
- [7] J. Shin, J. Cho, and S. Lee, "Please Touch Color: Tactile-Color Texture Design for The Visually Impaired," *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020.
- [8] N. A. Ibraheem, M. M. Hasan, R. Z. Khan, P. K. Mishra, "Understanding color models: a review," *ARPJ Journal of science and technology*, vol. 2, no. 3, pp. 265–275 2012.
- [9] J. E. Tyler and A. C. Hardy, "An Analysis of the Original Munsell Color System," *Journal of the Optical Society of America*, vol. 30, no. 12, p. 587, 1940.
- [10] R. Kitada, T. Kito, D. N. Saito, T. Kochiyama, M. Matsumura, N. Sadato, and S. J. Lederman, "Multisensory Activation of the Intraparietal Area When Classifying Grating Orientation: A Functional Magnetic Resonance Imaging Study," *Journal of Neuroscience*, vol. 26, no. 28, pp. 7491–7501, 2006.
- [11] M. A. Lawrence, R. Kitada, R. L. Klatzky, and S. J. Lederman, "Haptic Roughness Perception of Linear Gratings via Bare Finger or Rigid Probe," *Perception*, vol. 36, no. 4, pp. 547–557, 2007.
- [12] A. Pérez-Bellido, R. D. Pappal, and J. M. Yau, "Touch engages visual spatial contextual processing," *Scientific Reports*, vol. 8, no. 1, 2018.
- [13] K. Sathian and R. Stilla, "Cross-modal plasticity of tactile perception in blindness," *Restorative Neurology and Neuroscience*, vol. 28, no. 2, pp. 271–281, 2010.
- [14] NASA Task Load Index (NASA-TLX). [Online]. Available: <https://www.millisecond.com/download/library/NASATLX/>. [Accessed: 31-Jul-2020].
- [15] "SUS - A quick and dirty usability scale." [Online]. Available: <https://hell.meiert.org/core/pdf/sus.pdf>. [Accessed: 31-Jul-2020].
- [16] S. E. Palmer and K. B. Schloss, "An ecological valence theory of human color preference," *Proceedings of the National Academy of Sciences*, vol. 107, no. 19, pp. 8877–8882, 2010.
- [17] Website: <https://neilpatel.com/blog/color-psychology/>, Accessed on: 31-Jul-2020, [Online]
- [18] R. Rathee and P. Rajain, "Role Colour Plays in Influencing Consumer Behaviour," *International Research Journal of Business Studies*, vol. 12, no. 3, pp. 209–222, 2019.
- [19] S. G. Hart, "Nasa-task load index (NASA-TLX); 20 years later," *PsycEXTRA Dataset*, 2006.